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Statistical Learning and Large Data

Predicting Financial Distress From Orbis Data

A Machine Learning Approach

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Abstract

Can a firm's collapse be read directly from its balanche sheets? And can we identify structural determinants of financial distress years before the final breaking point? This paper answers affirmatively to both questions. Drawing on a stratified sample of over 30,000 firms from the Orbis repository, we develop a somewhat non-standard machine learning pipeline – hierarchical variable clustering, PCA-based dimensionality reduction, and LASSO penalized regression – that compresses a 199 variables feature space into just three structural predictors: solvency, profitability, and commercial scale. A parsimonious logistic model estimated on variables lagged two years prior to distress achieves around 80% out-of-sample accuracy and sensitivity, substantially outperforming the Altman Z-score benchmark. A Random Forest specification indirectly confirms the variable selection. The evidence suggests that financial distress is not sudden: it unfolds as time goes by, along these three main dimensions, making early detection feasible and, with some caution, reliable.

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Introduction

Predicting financial distress for firms and corporations is perhaps the most relevant and discussed topic in the empirical financial literature. The present report has a double aim: on the one hand, we want to verify whether it is possible to build a reasonably reliable predictive model capable of discriminating in advance between financially "solid" firms and corporations that have already embarked on the road to distress – intended as a state of insolvency, default, or outright liquidation – by using the accounting data made available by the Orbis (Bureau van Dijk) data repository; on the other hand, we want to understand whether the distress status is actually directly readable from a firm's balance sheets (that is, whether it emerges in the accounting stocks and flows as time goes by and the final date of default approaches). In order to explore these research questions, we extend to the Orbis data the contributions by Altman (1968) and Altman (1983) – who uses a very small number of balance sheet indexes – and, to this end, we employ a data-driven machine learning approach finalised at selecting the only most relevant variables in a very large and 'messy' real dataset.
The research outline is structured on different levels. On the purely predictive side, we try to correctly classify firms according to a bivariate outcome "active"/"distressed", aiming to obtain an acceptable error rate on a test set of randomly selected companies. On the descriptive/hermeneutic level, we search for the strongest predictors of this binary status, and compare them with the three main theoretical channels for predicting distress as identified (often *a priori*) by the financial literature – redditivity, solvency, and liquidity. At the very end of our work, we investigate non-parametric approaches (Random Forest) to determine if they do manage to achieve a systematic gain over linear methods, by effectively capturing non-linearities within a company's distress dynamics or mechanisms that are too complex to be accounted for by a standard supervised model.
On the practical side, the relevance of this study is immediate: banks, hedge funds, and investors all build models of this kind for credit scoring and risk management. Having a tool which, two years before financial disruption, classifies companies with a significantly higher-than-chance level of accuracy represents a useful screening device for these actors.

1 Dataset and Exploratory Data Analysis

The dataset is extracted from Orbis (BvD)[1], one of the top global repositories of balance sheet data; due to the immense number of firms and companies covered by

[1] https://orbis-r1.bvdinfo.com/version-20260316-5-2/Orbis/1/Companies/Search

the repository, for our analysis we employ a random sample stratified by the firm's status. Our initial sample contained 48,000 companies, equally divided between "active" (distress = 0) and "distressed" (distress = 1), where distress indicates one of the following three statuses in the Orbis anagraph: insolvent, bankrupt, or in liquidation.

The poor quality of Orbis data is well known (Bajgar et al., 2020) in the literature, especially the great amount of missing data due to incorrect registration or different accounting legislation across different parts of the globe. For sampling purposes, we imposed some minimum requirements on the firms to be randomly selected. This first screen checked whether: (i) firms were not "small" firms – we only focused on "medium", "large", and "very large" firms according to the BvD classification, as microcompanies tend to deposit shorter and more incomplete balance sheets and seem to be subject to rather different financial dynamics; (ii) firms could be recognized as possessing a documented financial history of at least 3 years – other companies were unfortunately significantly more affected by missing data; (iii) firms were required to have available in Orbis at least some core variables, i.e., those used to construct the Altman Z-score: total assets, total liabilities, sales revenues, EBIT margin, working capital, current and deferred liabilities, and net worth.

We also excluded *a priori* public authorities, states, and governments (whose balance sheets follow different rules and logics with respect to businesses) and companies that had no recent data, as for those firms no forecast on future distress might be possible for any investor whatsoever.

On the technical side, the dataset was imported into R through eight separate Excel downloads (four for active firms and four for distressed ones), each extracted in subsequent tranches from Orbis due to the export limitations imposed by the platform. The Excel files were read while correctly encoding NAs by specifying the Orbis-specific placeholder "n.d." and transforming every entry into character strings, a prudent choice aimed at avoiding implicit coercions that might silently produce NAs in the presence of inconsistent numeric formats or inadequate decimal separators. The single data frames were then bound together after having assigned to each firm the dependent dummy variable `distress` before the merge.

The first post-import check regarded the presence of duplicates, identified via the `BvD ID Number`, the unique identifier of each business within the repository. Duplicates could emerge because we sampled with replacement under the same extraction criteria; their removal is necessary to avoid the same observation contemporaneously influencing both the models' estimation and evaluation.

A second structural intervention focused on the removal of firms operating in the financial sectors (NACE codex 64, 65, 66: Financial service activities, Insurance, and related auxiliary activities). This exclusion was motivated by the radically differ-

ent wealth structure of financial intermediaries compared to firms belonging to the productive and manufacturing sectors – high leverage is physiological rather than pathological; assets are predominantly intangible, and liquidity metrics follow an utterly divergent logic. Including these firms in the estimation of the same model as industrial and commercial firms would introduce a systematic structural heterogeneity that would render the estimated coefficients useless and impossible to interpret economically.

1.1 NA Treatment

Handling missing values is a delicate methodological issue, for sure the most critical passage in this entire analysis' workflow. The Orbis repository presents a high heterogeneity in the coverage of balance sheet data and indicators: some variables – typically those computed by Orbis itself – are only available for subsets of geographic areas or Years; others are systematically unavailable for smaller firms or specific countries. After the work described in the previous section, we computed that approximately 16% of the whole entries in our dataset consisted of missing values.

A naive treatment of NAs (e.g., excluding any observation with at least a missing value) would produce a final dataset with drastically reduced dimension and, more importantly, unavoidably biased. Indeed, If the data is not missing at random (MaR), any filtering based on the number of NA for observation introduces some sort of sample selection bias, provided that the missingness status is even mildly associated with the dependent variable of interest (in our case, the distress status). This is something we must account for (Horton and Kleinman, 2007).

After careful consideration, we decided to employ an iterative procedure for the handling of missingness inside our dataset. In fact, we realized that any missing value simultaneously corrupts both variables and observations: big chunks of missing values for some similar, poorly registered columns also make the percentage of each row's missing values spike. Treating NAs by only focusing on rows or columns would risk eliminating ultimately informative data.

As a first step, we removed variables (8) with more than 50% NA[2], as an indicator missing for over half of the sample cannot reliably provide any kind of statistically robust information, and any estimate of its distribution would be based on too small a subsample for it to be representative. Then, after filtering for rows, we removed other 28 variables with more than 25% NA on the remaining observations.

Row elimination followed an analogous criterion, applied to observations rather

[2] We based the choice of the threshold for columns on the skip found in the list of emptiest variables as shown in the appendix in Tab. 5

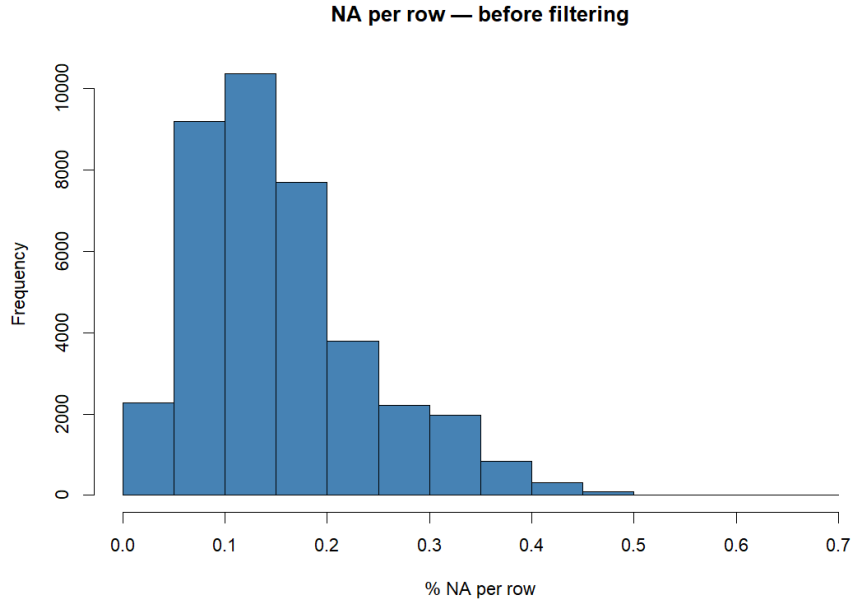


Fig. 1: NA pre-processing distribution

than to variables. For each comprised firms we computed the percentage of empty entries over the total variables available for the given phase of the procedure. Figure 1 shows the pre-processing distribution of NAs per row, on which we based our threshold choices for rows. Having too many NAs in a row would render the specific firm an artifact of synthetic data imputation rather than a genuine representation of the real company. We imposed subsequent thresholds of 20 and 10%, which brought the total residual NA percentage in the dataset down to a much more reasonable 3,1%.

Before proceeding with the eliminations, though, we performed a necessary missingness analysis in order to verify whether the elimination of observations was randomic or sytematic. This step is of utmost importance because, if eliminated firms present consistently different features than kept firms, the final sample is not representative of the targeted population and models' estimates may be biased. Through χ^2 independence tests, it emerged that the elimination probability is not randomic with respect to the geographic area ($\chi^2 = 1164.7, p < 0.001$), the firm's size ($\chi^2 = 73.7, p < 0.001$) and, crucially, the distress status ($\chi^2 = 169.96, p < 0.001$). Notably, firms coming from Centre and South America display an elimination rate of 61.3%, much higher than other parts of the world (and for this reason we drop all firms belonging to this zone: our analysis does not apply there) and distressed firms get eliminated 18.6% more than active ones. This is actually expected, as firms approaching distress usually tend to deposit incomplete balances whose values are later revised.

This evidence imposes caution in the interpretation of our results: our sample is

ill-conditioned due to data availability, and some geographic areas result underrepresented due to incompleteness. However, we should stress that this potential sample selection distorsion does not seem to show up in the out-of-sample predictions of our estimated models, which perform coherently on entities demographically similar to those included in the training set.
The final dataset, before feature engineering, comprises 30.977 firms: 17.575 active and 13.402 distressed, with a class ratio of approximately 1 distressed every 1.3 active – a slight imbalance which does not require aggressive rebalancing techniques.

1.2 Feature Engineering and Imputation

Once having obtained a reliable base for our dataset, the feature engineering phase pursued two distinct objectives: building economically interpretable variables from raw balance sheet numbers, and generating dynamic indicators capable of capturing financial disruption, which by definition anticipates default, as time goes by. We decided not to employ Orbis pre-computed indices, as they again reported a great amount of missing data and sometimes were present in duplicates with discording values – since it was not clear which would be the better choice for our analysis, we preferred to compute our own.
A first intervention concerned firms with non-positive total assets or liabilities: these observations were removed as they clearly were the result of an incorrect registration in the repository or of unfortunate accounting mistakes. We then built other variables such as the age of the firm[3], constructed as the difference between the year of the last available balance and the constitution year as recovered from the column `constitution date`, and the Altman Z-score ratios (Altman, 1983) for firms without market value of equity: X_1 = Working Capital / Total Assets (operating liquidity), X_2 = Net Worth / Total Assets (the best proxy we could find for retained earnings), X_3 = EBIT / Total Assets (redditivity of investments) and X_4 = Net Worth / Total Liabilities (solvency), X_5 = Sales Revenue / Total Assets.
The composite index would then be $Z'' = 0.717X_1 + 0.847X_2 + 3.107X_3 + 0.420X_4 + 0.998X_5$, with thresholds $Z'' >= 2.9$ for active firms, $1.23 <= Z'' < 2.90$ for the grey zone, and $Z'' < 1.23$ for distressed enterprises. Altman's model serves here both as a benchmark for the predictive performances of our machine learning models – which should NOT perform worse – and as a coherence validation for our data-driven approach: in other words, did we effectively retrieve approximately the same dimensions using only information coming from the data, i.e., without *a priori* reasonings?

[3] Age is a relevant proxy for a firm's financial solidity: younger firms typically present higher default rates, whereas very mature enterprises can suffer from structural rigidity.

We also performed winsorization at 1st and 99th percentiles in order to treat outliers, and built dynamic variables up to the third lag for each relevant predictor as first temporal differences. The separation between training and testing set was done prior to any imputation operation, with a 70/30 separation stratified with respect to the class in order to preserve balance.
Imputation parameters (mean and standard deviation for each variable) were computed solely on the training set in order to avoid data leakage. We conducted imputation using a simple mean-plus-noise method: missing values have been substituted with $\hat{\mu} + \epsilon_i$, where $\hat{\mu}$ is the variable's mean estimated on training data and $\epsilon_i \sim \mathcal{N}(0, \hat{\sigma}^2)$ is a gaussian error term with the same variance as the sample variance of the considered column. The same parameter set $(\hat{\mu}_j, \hat{\sigma}_j)$ has then been applied to each column j of the test set. We also tested different approaches as the KNN imputation, but ended up choosing the simpler method. Descriptive statistics of the final training dataset obtained after our cleaning procedure are displayed in the Appendix.

1.3 K-Means Clustering

We perform K-Means Clustering to investigate the underlying structure of the dataset. For computational reasons, we first operate on a well-balanced subset of 7,000 firms, verifying that the results generalise to the full dataset.

We find the optimal number of clusters maximising silhouette width to be 2; since 2 is also the number of categories of interest in our study, namely *active* and *distressed*, this promising result suggests that unsupervised classification of firms may indeed be feasible. However, when running a 2-Means Clustering algorithm on the entire scaled training dataset, we obtain a very small cluster of 22 large corporations, which clearly constitute outliers and are therefore removed from the analysis.

Even after removing these 22 firms, 2-Means Clustering still partitions the training dataset into a large cluster of heterogeneous firms and a small cluster of 474 very large companies. Since the dataset contains both ratio variables and variables expressed in absolute terms, this size-driven partition is likely attributable to the latter: despite standard scaling, absolute balance-sheet quantities retain a structural dispersion that systematically separates very large corporations from the rest of the sample.

As a robustness check, we normalise all absolute variables by the firm's total assets in the corresponding year, effectively expressing every balance-sheet quantity as a fraction of firm size. Running 2-Means Clustering on this size-adjusted dataset

yields a small cluster of 3 outliers, which are removed; the remaining partition is summarised in Tab. 1. The two clusters now exhibit a markedly different composition: Cluster 1 is predominantly active (78.8%), while Cluster 2 contains a majority of distressed firms (56.1%). The association is highly significant ($\chi^2 = 2448.4$, $p < 0.001$), confirming that, once size is controlled for, K-Means recovers a partition that is meaningfully aligned with the distress status.

Tab. 1: 2-Means Clustering on size-normalised data, by distress status

Cluster	Active	Distressed	Total	% Distressed
1	6,056	1,627	7,683	21.2%
2	6,010	7,695	13,705	56.1%

$\chi^2 = 2448.4$, $p < 0.001$

This is an encouraging result for an entirely unsupervised procedure, even if not sufficient for a sharp classification. For this reason, we choose to proceed with a supervised classification method.

2 Dimensionality Reduction

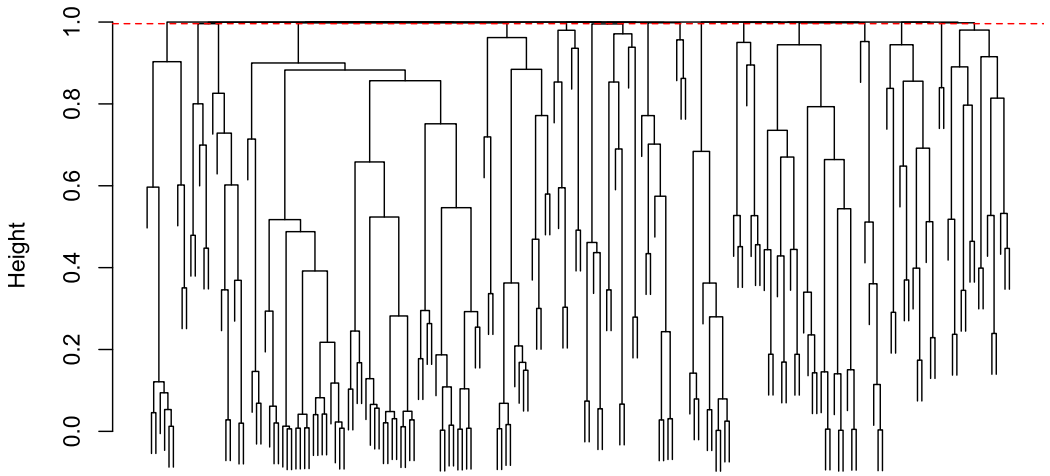
Considering the high dimensionality of the feature space, consisting in 199 variables, we apply multiple Statistical Learning techniques to reduce it. The pipeline is (i) Clustering the variables, (ii) summarizing the clusters by selecting the variables with the highest loading per Principal Component up to 70% cumulated explained variance for each cluster, (iii) selecting the most powerful variables with a LASSO.

2.1 Hierarchical Clustering of Variables

We first perform an agglomerative hierarchical clustering of the quantitative variables, using $1 - |r|$ as a distance metrics (being r the Pearson correlation coefficient).The aggregation follows the complete linkage criterion, which calculates the distance between two groups using their two most distant variables. The resulting dendrogram is provided in Fig. 2.

The choice to use the absolute correlation as a distance metric instead of its square is justified by the resulting partition. In fact, over a similar height, using $|r|$ favored a stronger aggregation of the variables, yielding fewer distinct clusters and thus facilitating a more effective subsequent summarization step.

Fig. 2: Dendrogram of the hierarchically clustered variables with complete linkage



By cutting the dendrogram at a height of 0.996003, we partition the feature space into 15 distinct clusters.

This choice implies that the maximum distance between any two variables within the same group is less than or equal to 0.996003. As a direct conclusion, even the two most distant variables in a cluster maintain a minimum absolute correlation of $|r| = 1 - 0.996003 = 0.003997$. Within each cluster, this high threshold allows the algorithm to be permissive enough to aggregate variables that are connected through a sequence of intermediate steps. At the same time, however, it imposes a definitive boundary that keeps highly divergent variables strictly apart, ensuring that elements without any meaningful direct or mediated statistical link are never forced into the same group. The conclusion here is that the boundaries of these 15 macro-groups have reached a point of near-total mutual repulsion. By adopting such a high threshold, we ensure that the algorithm stops merging branches only when the external extremes of the groups become completely incompatible, ensuring that the 15 blocks represent truly distinct information areas.

Concretely, choosing such a high threshold prevents the variables from splitting into too many small and isolated micro-groups. Since balance-sheet data are highly interconnected, moreover including lagged observations of the same variable, this approach allows variables from different financial areas to stay in the same cluster if they are tied together by intermediate bridge variables. The ultimate conclusion of this preprocessing step is that we obtain a clean, entirely data-driven taxonomy that successfully organizes the entire dataset into 15 independent blocks of variance with enough dimensionality to obtain significant results when analysing the Principal Components of each cluster.

2.2 Principal Components

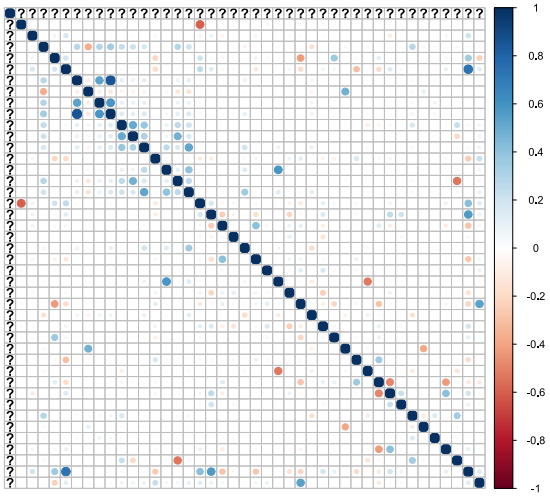
To effectively reduce the dimensionality of our dataset, we develop a feature selection pipeline within each of the 15 previously identified clusters. First, a Principal Component Analysis (PCA) is performed individually on each cluster to capture its internal variance. We then retain a subset of principal components up to the point where the cumulative explained variance within that specific cluster reaches at least 70%.

For each retained component, we identify and select the underlying original variable that exhibits the highest absolute loading, ensuring that we capture the feature with the strongest representative power for that dimension of variance.

However, to maximize the economic interpretability of the final model, when absolute loadings are substantially close (i.e., within a negligible numerical tolerance), if the variable with the highest absolute loading represents a flow or a lagged difference, but a non-flow (level) variable exhibits a nearly identical loading, we prioritize the non-flow variable. This trade-off deliberately favors financial meaningfulness over a marginal, statistically negligible gain in explained variance.

This systematic approach allows us to condense the original high-dimensional feature space into a highly informative and economically sound subset of 41 quantitative variables.

Fig. 3: Full feature space corrplot



It is possible to provide an economic interpretation to many of the clusters created, in light of the variables selected by the PCA belonging to them. Indeed, several clusters naturally group together variables that share a common economic meaning, suggesting that the PCA is able to capture meaningful underlying dimensions of firm performance. The variables selected from clusters 2, 5, 10, and 11 all refer to measures of profitability and returns, whereas clusters 3, 4, 6, and 13 consist of various measures of capital stock and assets. Finally, cluster 8 is solely composed of leverage indexes, cluster 9 of liquidity indexes, and cluster 12 of solvency indexes.

We further acknowledge that there is no perfect orthogonality between the variables so selected, mostly due to the fact that many variables hail from the same cluster. This can be seen in the correlation plot shown in Fig. 3, computed over the 41 selected quantitative variables and the age of the firm. However, most predictors are indeed orthogonal, and VIFs are lower than 5, so that we can safely procede with a LASSO-driven filtering of the final sparse variable selection.

2.3 LASSO Binomial Regression

To further reduce the dimensionality of the dataset, we estimate a binomial LASSO regression of the probability of financial distress on the 49 selected predictors. The LASSO penalty shrinks the coefficients of irrelevant variables exactly to zero, effectively performing a variable selection. The penalization parameter λ is chosen as the largest whose error lies within standard error of the λ effectively minimizing cross-validation loss. We are able to follow this 1 standard error rule because the cross-validation plot, displayed in Fig. 4 shows that the increase in binomial deviance is minimal, wherease the gain in number of predictors penalized is very high, since it allows to transition from 23 retained variables to only 9.

The shrinkage is documented in Fig. 5, which illustrates the LASSO coefficient paths as a function of λ. The 8 most resilient coefficients are all negative, whereas the 9th and last (by following the 1se rule) coefficient, namely the Liquidity Ratio t_0, is positive. A possible explanation may be that a distressed firm possesses more unproductive cash, and in this sense the Liquidity Ratio is a signal and not a cause of distress.

Fig. 4: LASSO cross-validation plot

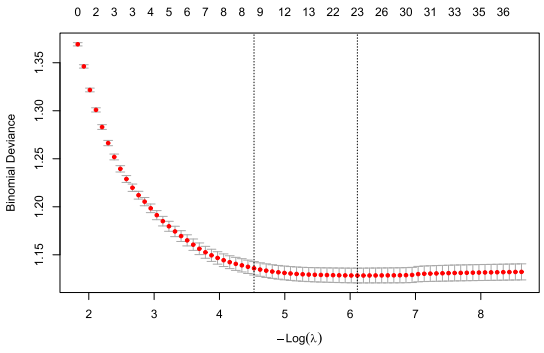
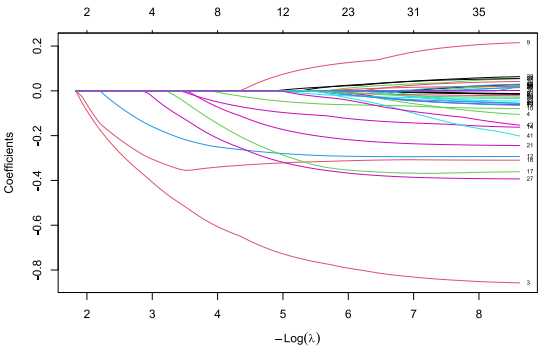


Fig. 5: LASSO coefficient paths plot



Moreover, the coefficient paths plot allows to trace the most important variables under penalization. They are, in decreasing order of magnitude, Solvency Ratio t_{-2}, ROA t_0, and EBIT t_0. As a matter of fact, last two coefficients to be penalized, i.e. Solvency Ratio t_{-2} and ROA t_0, happen to be so almost simultaneously, at a

nearly identical log-lambda threshold, indicating that both profitability and long-term structural solvency act as twin pillars in signaling financial distress, and thus validating the financial intuition that a company's collapse is rarely a matter of a single isolated metric, but rather a synchronized failure of both current earning capacity and long-term capital stability.

3 Logistic Regression

Although the binomial LASSO is excellent for variable selection, its regularization mechanism shrinks coefficients toward zero, which underestimates the true economic impact of predictors on financial distress. To retrieve unbiased and interpretable parameters, we adopt a two-stage post-LASSO strategy by running a standard Logistic Regression restricted only to the 9 quantitative variables selected by the LASSO. To isolate genuine risk factors, the baseline model was initially augmented with categorical controls for systemic macroeconomic shocks via year dummies, firm size, geographical origin, age, and corporate complexity measured by the number of subsidiaries.

However, only year dummies and geographical area resulted statistically significant. Structural controls such as firm size, age, and the number of subsidiaries showed no significant explanatory power, suggesting that their demographic information is already fully captured by the 9 quantitative balance-sheet indicators. Conversely, temporal and spatial factors exert an autonomous, non-negligible impact on firm survival. Consequently, we drop the non-significant controls to preserve parsimony. The final Model 1 displayed in Tab. 2 expresses the probability of financial distress solely as a function of the 9 quantitative predictors, controlling for year and geographical area.

All signs observed in the LASSO coefficients paths are confirmed, with Liquidity Ratio t_0 being the only variable with positive coefficient. Moreover, all coefficients are statistically significant.

In line with the literature Altman (1983) and to prevent look-ahead bias, Model 2 is restricted exclusively to variables with a t_{-2} lag. Although contemporaneous variables exhibit strong correlative power, they reflect an ongoing crisis (e.g., cash-hoarding in the case of liquidity) rather than anticipating it. Feature selection was intentionally applied to the unrestricted pool of all lags (t_0, t_{-1}, t_{-2}), rather than re-running LASSO solely on t_{-2} variables. While contemporaneous metrics (t_0) hold an inherent statistical advantage due to their synchronicity with distress, the survival of t_{-2} predictors in a competitive feature space serves as a rigorous empirical stress test. This strategy validates the selected regressors as highly resilient leading indicators

whose predictive power is not overshadowed by the acute, short-term signals of the pre-default phase.

Tab. 2: Logistic Regressions

Dependent variable:	**Distress** Model 1	**Distress** Model 2
Solvency Ratio t_{-2}	-0.026***	-0.0219***
Solvency Ratio t_{-1}	-0.022***	
Solvency Ratio t_0	-0.009***	
Liquidity Ratio t_0	0.059***	
EBIT Margin t_{-2}	-0.007***	-0.0154***
EBIT Margin t_0	-0.012***	
ROE (Gross) t_0	-0.002***	
ROA (Gross) t_0	-0.020***	
Sales Revenue t_{-2}	-0.000002***	-0.00000217***
Constant	-2.584**	-2.629***

Notes: $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$

3.1 Comparing Models

To rigorously assess the predictive power of the engineered features and evaluate the effectiveness of our dimensionality reduction pipeline, we benchmark several logistic specifications on the test set. This out-of-sample (OOS) validation prevents overestimating the models' performance and ensures structural robustness. The performance metrics are summarized in Tab. 3. A crucial finding emerges when comparing the unrestricted model containing all 49 PC-selected variables against Model 1, which restricts the pool to the 9 quantitative predictors selected via LASSO binomial regression. At a classification threshold of 0.5, both models exhibit nearly identical predictive performance. This empirical equivalence provides powerful justification for our pipeline: filtering the feature space through the LASSO penalty successfully discards redundant noise and retains the entire statistical signal while achieving a massive reduction in model complexity.

For a financial institution, failing to detect an impending default (a false negative) is substantially more expensive than misclassifying a solid firm as distressed (a false positive). Consequently, we implement a progressive optimization of the decision boundary, lowering the probability threshold to 0.4. As expected, shifting the threshold to 0.4 boosts the Sensitivity from 76.16% to 82.76%, successfully capturing a larger portion of distressed corporate entities at the cost of a minor drop in Specificity (83.95%). The progressive optimization of the probability threshold from 0.5 to 0.4 highlights the structural trade-off between baseline accuracy and the

model's sensitivity in correctly identifying distressed corporate entities.

Finally, Model 2 addresses look-ahead bias by restricting the feature pool exclusively to twice-lagged variables (t_{-2}). When evaluating this strictly predictive model (at a 0.4 threshold), the accuracy remains remarkably resilient at 79.67%, maintaining a high sensitivity of 80.18%. This proves that the survival of t_{-2} predictors in the competitive LASSO stage acts as a rigorous stress test : these features serve as resilient leading indicators capable of capturing structural insolvency two years before the acute crisis phase manifests. Conversely, removing temporal controls completely (the "t_{-2} without years" specification) causes performance to degrade sharply to an accuracy of 71.87%, confirming that macroeconomic temporal shocks exert a non-negligible, autonomous impact on firm survival.

Tab. 3: Out-of-Sample predictive performace across different model specifications

Model Specification	**Thr.**	**Accuracy**	**Sensitivity**	**Specificity**
ALL 49 selected variables	0.5	84.40%	75.83%	90.89%
9 LASSO-selected variables	0.5	84.12%	76.16%	90.16%
9 LASSO variables (FT)	0.4	83.44%	82.76%	83.95%
Model 2 (only t_{-2})	0.4	79.67%	80.18%	79.29%
Model 2 (t_{-2}), without years	0.5	71.87%	71.49%	72.17%
Altman Z (Grey is Distress)		56.09%	71.19%	44.28%
Altman Z (Grey is Active)		60.67%	32.8%	82.49%
Altman Z (Without Grey)		63.59%	53.24%	71.66%

Note: All performance metrics are evaluated out-of-sample on the test dataset. The Altman Z-Score without Grey Zone is tested on a structurally smaller set.

The most surprising result is indeed the sharp OOS performance of Model 2, which with only 3 lagged variables is still able to provide >80% accuracy and sensitivity and specificity. This remarkable efficiency confirms that structural solvency, operational profitability, and commercial scale act as the primary, intertwined dimensions of long-term corporate health. By effectively capturing these three distinct economic channels, Model 2 proves that a highly parsimonious framework can synthesize the entire predictive signal without requiring short-term accounting noise. This establishes a robust baseline, demonstrating that a firm's collapse is fundamentally rooted in these core structural pillars.

Finally, it is clear how all our models substantially outperform Altman's Z-Score across every performance dimension. The Z-Score benchmark is further complicated by the ambiguity of its Grey Zone, whose classification introduces structural uncertainty: depending on whether Grey Zone firms are treated as distressed or active, Sensitivity and Specificity swing dramatically, rendering the metric unreliable as a standalone decision tool. Even when Grey Zone firms are excluded entirely, reduc-

ing ambiguity at the cost of a structurally smaller evaluation set, the Z-Score still falls short of Model 2 without temporal effects on all reported statistics. This is a remarkably strong result: our parsimonious logistic model, relying on only three twice-lagged predictors and no short-term accounting information, consistently outperforms Altman's benchmark.

3.2 Partition Plot

Before testing non-parametric approaches, we check whether the decision boundary between active and distressed enterprises presents a simple geometric structure in a bidimensional feature space. To this end, we build two partition plots using Quadratic Discriminant Analysis (QDA) (James et al., 2013).
For the visualization, we use the two most powerful explanatory variables as selected by the LASSO penalised regression – Solvency Ratio and ROA –, as well as a linear combination of all the variables selected by the LASSO by taking their first two principal components. Both plots show that there exists no visible split between the two classes in two dimensions: the overlapping zone where safe and disrupted firms are mixed together is substantial.
The choice of employing QDA rather than its linear alternative (LDA) is motivated by two reasonings. First, the assumption of equal variance-covariance matrices between classes required by LDA seems fragile in our context: distressed firms tend to present greater dispersion in the financial variables than active businesses, making the homoskedasticity hypothesis structurally implausible. Second, the high sample size is amply sufficient to separately estimate the distinct variances of the two cathegories, without incurring in the instability problems typical of a small-sample QDA. In any case, the bidimensional analysis yields deluding results, effectively confirming the high-dimensional nature of the financial distress problem and prompting our analysis towards a more non-parametric approach, which we illustrate briefly in the next session.

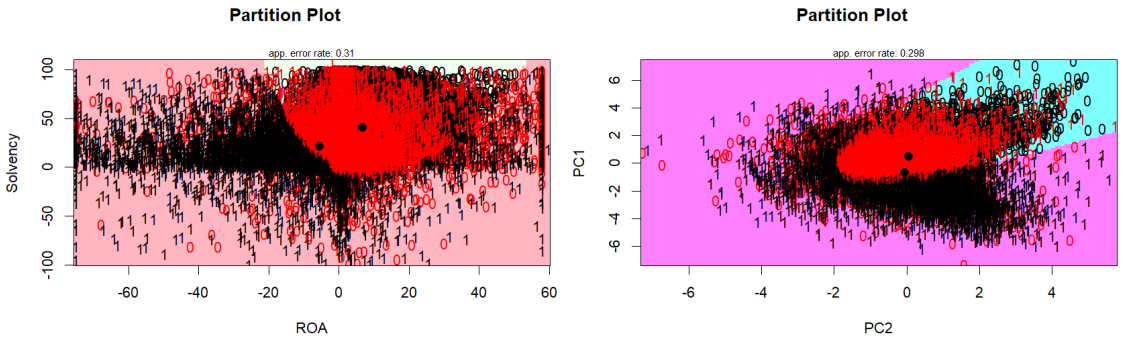


Fig. 6: Partition plots as resulting from QDA.

3.3 Random Forest

The Random Forest is an ensemble non-parametric approach based on the aggregation of a great number of decision trees, each trained on a bootstrap subsample of the observations and a random subsample of variables at each node (James et al., 2013). The final prediction is the majority vote across all trees.
In the present work, the Random Forest was trained on 500 trees and 7 candidate variables at each split, a value chosen in accordance to the empirical rule of $\sqrt{p}$ for binary classification given that we had 41 total variables as selected by the PCA and which we used to identify non linearities. The relative importance of the variables is measured through the Mean Decrease Accuracy criterion: for each variable, we compute how the model's accuracy decreases when that specific variable is randomly permutated in the out-of-bag observations, effectively neutralising its predictive power.
We estimate two Random Forest classifiers: Model 1 includes the fiscal year as a categorical predictor alongside all financial covariates, while Model 2 excludes it, serving as a robustness check against time-confounding. Both models are trained on the same 70/30 stratified train–test split used throughout the analysis, with classification thresholds selected via the Youden index to balance sensitivity and specificity. M1 achieves an AUC of 0.934 on the held-out test set, with accuracy of 86.5%, sensitivity of 84.6%, and specificity of 88.0% at the optimal threshold. M2, estimated without the fiscal year variable, yields an AUC of 0.895, confirming that high importance of the macroeconomic trend in the years covered by the sample (probably due to the global pandemic in the early 2020s). The AUC gap of approximately 3.9 percentage points suggests that fiscal year captures meaningful cyclical variations.
Feature importance plots based on Mean Decrease Accuracy confirm the economic interpretability of the models. Across both specifications, the leading financial predictors are the Solvency Ratio (T−2), Earnings Before Tax (T0), ROA (T0), ROE (T0), and EBIT Margin — coherent with our previous analyses. The prominence of lagged variables (T−1, T−2) corroborates the view that financial deterioration is gradual and detectable before the distress event materialises. In M1, Fiscal Year and Geographic Area rank first and second by importance, which motivates the M2 specification: once the temporal variable is removed, the purely financial predictors retain substantial discriminatory power, underscoring the validity of the underlying cross-sectional signal.

Taken together, the Random Forest results serve as a non-parametric upper bound on predictive performance for this feature set, and corroborates the variable selection operated by the LASSO step, since the most important features in the Random Forest largely coincide with those retained by the penalised regression.

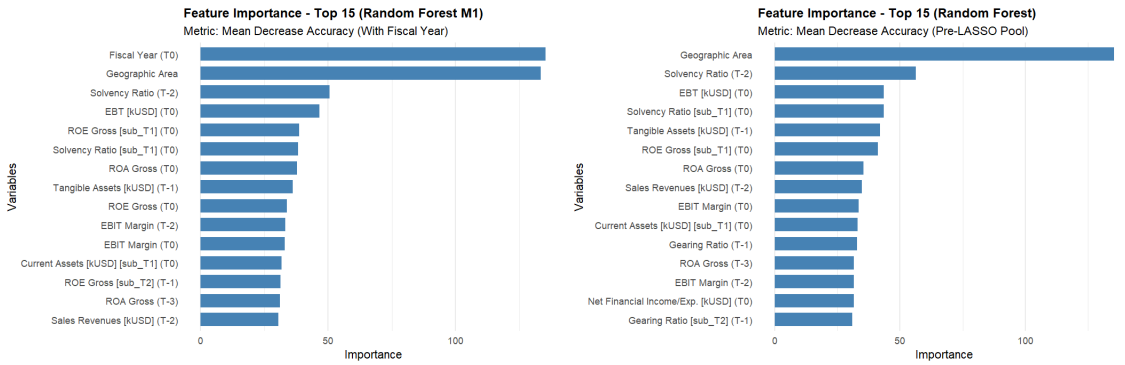


Fig. 7: Variable Importance plots for the two specifications

Conclusion

This work has developed and validated a data-driven statistical learning workflow for the prediction of corporate financial distress, employing balance data sheets available on Orbis depository, with remarkable OOS performances of the resulting models.

The Orbis dataset is notoriously full of missing values and we rigorously approached NAs with an iterative process consiting in columns-rows-columns-rows, effectively adopting a very conservative way to reduce as little rows as possible. At the end of the procedure, only 3.1% of all observations were missing. At this point, we removed all firms with unreasonable, non-positive assets or liabilities and exluded outliers through winsorization. We finally imputed NAs with the variable's mean plus gaussian noise with the same variance as the variable variance, computing all parameters solely on the training set to avoid data leakage.

The dimensionality reduction pipeline, combining hierarchical clustering of variables, PCA-based feature selection, and LASSO binomial regression, effectively reduced a feature space of 199 variables into only 9 core indicators, with no measurable loss in out-of-sample classification performance relative to the full 41-variable specification. Furthermore, a very parsimonious model for a two-years-in-advance alert on financial distress, employing only 3 predictors mapping three main dimensions of business life: solvency, redditivity, and commercial scale. Finally, we adopted a non-parametric classification approach to address eventual non-linearities in the datast, through a Random Forest. This method outperforms Logistic Regressions, at the expense of parameters' direct interpretability.

Further research involving more robust handling techniques of observations corrupted by too many missing values may allow for a more precise and reliable model, and to extend the population of firms to excluded markets such as South and Central America.

Replication Materials

The R codes employed for the analysis presented in this work are available at

`https://github.com/paolochiabrera/Financial--distress`.

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A Appendix

Tab. 4: Summary statistics

Variable	N	Mean	SD	Q25	Median	Q75
Total Value of Production (Thous. USD)	21225	27109.20	142354.90	1259.95	2362.47	6100.62
Profit/Loss Before Tax (Thous. USD)	21225	842.44	24853.63	-67.57	37.89	245.77
Net Income (Thous. USD)	21225	453.09	21445.17	-66.37	26.91	193.60
Total Assets (Thous. USD)	21225	36361.38	366878.85	1043.06	2487.40	6504.94
Shareholders' Equity (Thous. USD)	21225	13448.05	152862.98	94.80	556.01	2124.24
Current Ratio	21225	2.87	6.22	0.93	1.37	2.41
Profit Margin (%)	21225	0.99	20.25	-2.63	1.70	7.36
ROE - Gross (%)	21225	3.94	82.89	-2.40	9.68	32.48
Intangible Assets (Thous. USD)	21225	2636.49	59141.35	0.00	0.00	29.95
Tangible Assets (Thous. USD)	21225	11255.86	179288.99	31.08	226.17	1301.59
Inventories (Thous. USD)	21225	4444.57	88583.44	0.00	101.41	703.57
Accounts Receivable (Thous. USD)	21225	4712.63	57971.48	54.02	322.32	1092.85
Cash and Cash Equivalents (Thous. USD)	21225	3007.17	30253.42	25.73	130.91	545.27
Share Capital (Thous. USD)	21225	3079.41	51640.87	10.39	58.84	306.12
Non-Current Liabilities (Thous. USD)	21225	9651.31	141626.61	16.10	234.26	1148.75
Current Liabilities (Thous. USD)	21225	13259.35	129008.00	362.59	961.17	2835.87
Sales Revenues (Thous. USD)	21225	26235.46	137518.52	1203.22	2311.67	5854.01
Cost of Goods Sold (Thous. USD)	21225	17486.77	99240.53	145.89	1160.03	4975.94
Contribution Margin (Thous. USD)	21225	9989.45	69600.68	356.66	1022.36	2948.20

Tab. 4: Summary statistics *(continued)*

Variable	N	Mean	SD	Q25	Median	Q75
Financial Income/Expenses (Thous. USD)	21225	-271.83	7754.19	-53.45	-7.43	0.00
Current Assets (Thous. USD)	21225	16490.61	172330.16	596.79	1429.45	3771.98
Working Capital (Thous. USD)	21225	4732.71	79973.89	0.00	328.78	1623.47
ROA - Gross (%)	21225	1.24	19.42	-2.58	1.95	8.84
EBITDA Margin (%)	21225	6.16	22.02	-0.80	4.53	12.90
EBIT Margin (%)	21225	2.39	20.54	-1.82	2.57	8.46
Gearing Ratio (%)	21225	115.88	176.40	3.73	48.62	171.03
Solvency Ratio (%)	21225	31.25	35.12	9.08	29.43	56.58
Liquidity Ratio	21225	2.13	3.87	0.57	1.06	1.95
Registered Subsidiaries	21225	0.82	4.96	0.00	0.00	0.00
Establishment Year	20912	2002.86	15.38	1995.00	2007.00	2015.00
Company Age	20909	20.70	15.49	9.00	17.00	29.00
Total Liabilities (Thous. USD)	21225	22910.67	238614.85	586.19	1566.42	4352.66
EBIT (Thous. USD)	21225	1117.29	25005.40	-49.23	56.82	281.91
X1	21225	0.20	0.32	0.01	0.20	0.41
X2	21225	0.32	0.31	0.11	0.28	0.53
X3	21225	0.05	0.15	0.00	0.04	0.10
X4	21225	1.57	4.47	0.12	0.39	1.11
X5	21225	1.78	4.44	0.61	1.27	2.20
Z-Score	21225	2.99	3.02	1.28	2.32	3.78

Tab. 5: Summary of Missing Data per Variable (t_0 only)

Variable Name	**Missing Values (%)**
Debts (k USD)	100.0%
R&D Costs (k USD)	94.7%
Solvency Ratio (Liquidity-based)	49.1%
Net Working Capital (k USD)	37.0%
Return on Capital Employed (ROCE) - Gross	36.8%
Cash Flow (k USD)	34.7%
Interest Coverage Ratio	33.8%
Number of Employees	33.1%
Gearing Ratio	25.6%
EBITDA Margin	21.7%
Return on Equity (ROE) - Gross	21.5%
Cost of Goods Sold (COGS) (k USD)	20.0%
Gross Margin (k USD)	17.9%
Working Capital (k USD)	16.3%
Profit Margin	8.5%
EBIT Margin	7.3%
Solvency Ratio (Asset-based)	5.9%
Cash & Cash Equivalents (k USD)	5.5%
Return on Assets (ROA) - Gross	4.1%
BvD Sectors	2.2%
Liquidity Ratio	1.8%
NACE Rev. 2 (Primary Sector/Codes)	1.8%
Current Ratio	1.8%
Intangible Assets (k USD)	1.5%

Continues below

Table 5 – (continued)

Variable Name	Missing Values (%)
Date of Incorporation	1.3%
Tangible Assets (k USD)	1.2%
Non-current Liabilities (k USD)	0.8%

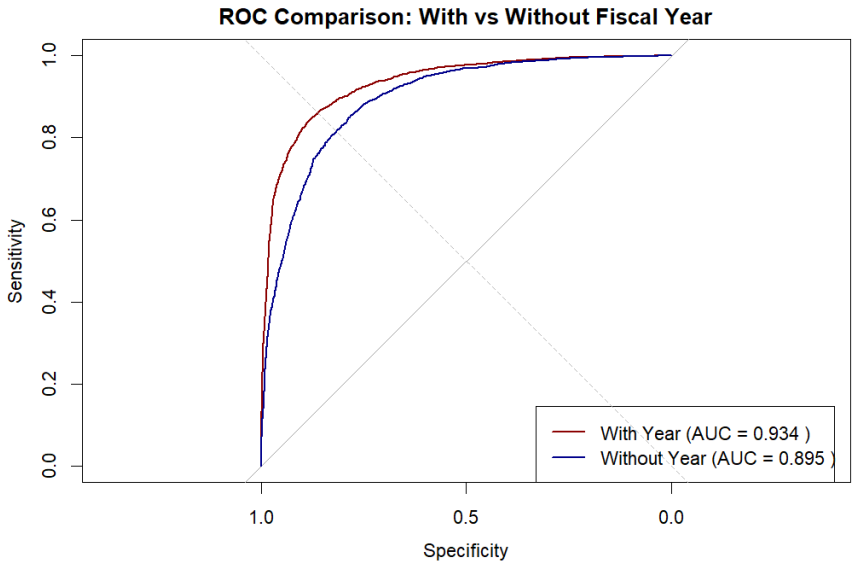


Fig. 8: ROC curves for the two Random Forest specifications.